



New AI tool turns low-pixel faces into realistic images

Researchers at Duke University in Durham, North Carolina have developed an artificial intelligence (AI) tool that can turn blurry and unrecognisable images of people's faces into perfect computer-generated portraits. The method, called PULSE, searches through AI-generated examples of high-resolution faces to match the ones that look similar to the input image when compressed to the same size.

"While the researchers focused on faces as a proof of concept, the same technique could, in theory, take low-resolution shots of almost anything and create sharp, realistic-looking pictures, with applications ranging from medicine and microscopy to astronomy and satellite imagery," says co-author Sachit Menon from the Duke University.

Instead of starting with a low-resolution image and slowly adding detail, PULSE traverses a high-resolution natural image manifold, searching for images that downscale to the original low-resolution image. The team used a tool in machine learning called 'generative adversarial network' or GAN, which are two neural networks trained on the same data set of photos.

One network comes up with AI-created human faces that mimic the ones it was trained on, while the other takes this output and decides if it is convincing enough to be mistaken for the real thing. The system can convert a 16x16 pixel image of a face to 1024 x 1024 pixels in a few seconds. Details such as pores, wrinkles, and wisps of hair that are imperceptible in the low-pixel photos become crisp and clear in the computer-generated versions.



Representational image

Telcos increase vigilance as government warns of phishing attacks

Telecom companies have increased their vigilance of networks and systems following the government's recent warning on large-scale phishing attacks in the coming days against individuals and businesses, where attackers may use Covid-19 as bait.

In a security advisory to its business-to-business customers, Airtel says it is monitoring the situation on a real-time basis and has upgraded the security operations centre to a higher threat-level status and promised to send daily dashboards to customers.

Citing various reports, the company warns that cyber-attacks can be initiated by exploiting various vectors such as e-mail/phishing attack, suspicious webpages, internet links, unpatched/un-updated systems connected on the network, or identity theft (poor ID management). A source from Jio says that the company has raised its vigilance levels and is keeping a close watch on traffic.

Recently, India's cybersecurity nodal agency, CERT-In issued an advisory warning of the potential phishing attacks that could impersonate government agencies, departments and trade bodies which have been tasked to

oversee disbursement of government fiscal aid. The attackers are expected to send malicious e-mails under the pretext of local authorities that are in charge of dispensing government-funded Covid-19 support initiatives.



Telecom companies have increased vigilance following the government's recent warning (Credit: www.pixabay.com)